

Event Boundaries and Memory in the Mnemonic Similarity Task

BRSM 2026 — Descriptive and Inferential Analysis Report

MST Simpletons

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1 Introduction

Human experience is not a continuous stream — the brain automatically segments ongoing activity into discrete *events*. An **event boundary** occurs whenever the perceptual or situational context changes, such as moving between rooms or a scene change in a film (Zacks, Speer, Swallow, Braver, & Reynolds, 2007). These boundaries critically determine what is encoded into memory. Two complementary effects have been documented. The **Blind Spot effect**: items immediately *after* a boundary are remembered less accurately, because the cognitive system is occupied updating its internal event model (Swallow, Zacks, & Abrams, 2009; DuBrow & Davachi, 2014). The **Snapshot effect**: items immediately *before* a boundary may be encoded with enhanced fidelity, as the impending context shift signals the brain to stabilise the current representation (Heusser, Ezzyat, Shiff, & Davachi, 2018).

The **Mnemonic Similarity Task** (MST; Stark, Yassa, Lacy, & Stark, 2013; Stark, Kirwan, & Stark, 2019) is ideally suited to investigate these effects. During encoding, participants study object images; during test, they classify each image as “Old” (target), “Similar” (lure), or “New” (foil). The MST yields two orthogonal metrics: **Recognition Memory (REC)**, capturing old/new discrimination, and the **Lure Discrimination Index (LDI)**, capturing hippocampal pattern separation ability. Following Morse, Duff, Abel, and Johnson (2023), we embedded event boundaries into the encoding sequence by interleaving scene-change images between blocks of objects. A total of **159 participants** were tested across three groups varying in encoding task complexity: **Item Only** ($n = 56$), **Task Only** ($n = 53$), and **Both Item & Task** ($n = 50$ – 52). Items at Position 1 (post-boundary), Position 4 (mid-event baseline), and Position 7 (pre-boundary) within each block were compared. Three directional hypotheses were tested:

- **H1 — Blind Spot**: Post-boundary items show lower REC than mid-event items.
- **H2 — Snapshot**: Pre-boundary items show higher LDI than mid-event items.
- **H3 — Speed Bump**: Post-boundary items elicit slower retrieval RT than mid-event items.

2 Methods

2.1 Experimental Design

The sample comprised 159 participants assigned to one of three groups based on the encoding task performed. The **Item Only** group ($n = 56$) indicated whether each object was indoor/outdoor. The **Task Only** group ($n = 53$) performed a categorisation task (e.g., “Does it fit in a shoebox?”). The **Both Item & Task** group ($n = 50$ – 52) performed both tasks simultaneously. During encoding, participants viewed 280 object images in 40 blocks of 7 items, with a scene change and task shift at each block transition serving as the event boundary. Items were classified by position: **post-boundary** (Position 1), **mid-event** (Position 4, baseline), and **pre-boundary** (Position 7). During the subsequent test phase, participants classified 150 images — targets, lures (across five similarity bins, Bin 1 = most similar), and foils — as “Old”, “Similar”, or “New”.

2.2 Variables Computed

Two bias-corrected memory indices were computed per participant at each boundary position:

$$\begin{aligned}\text{REC} &= P(\text{“Old”} \mid \text{Target}) - P(\text{“Old”} \mid \text{Foil}) \\ \text{LDI} &= P(\text{“Similar”} \mid \text{Lure}) - P(\text{“Similar”} \mid \text{Foil})\end{aligned}$$

REC captures recognition accuracy; LDI captures pattern separation ability. Subtracting the foil false-alarm rate corrects for individual response bias — a participant who responds “Old” to everything achieves a high hit rate but zero genuine discrimination. This subtraction ensures that both metrics reflect true memory rather than response tendency. Retrieval reaction time (RT) was also recorded for each test trial.

2.3 Statistical Analysis

Descriptive statistics. Means, standard deviations, medians, and interquartile ranges were computed for REC and LDI at each boundary position within each group. Distributions were visualised with violin plots overlaid with box plots and group means. Line plots with ± 1 SE error bars illustrated positional trends. A lure bin accuracy curve — $P(\text{“Similar”} \mid \text{Lure})$ across five similarity bins — validated the MST: a monotonically increasing curve from Bin 1 to Bin 5 confirms genuine pattern separation.

Normality assessment. Before applying parametric tests, normality was assessed in each of the nine group $\times$ position cells using the Shapiro–Wilk test ($p < 0.05$ indicates departure) and QQ plots. Where departures were found, non-parametric alternatives were run in parallel.

Inferential tests. The primary tests were **paired-samples t -tests** — appropriate because boundary position is a within-subject manipulation, with each participant contributing data

at all three positions. This design uses participants as their own controls, reducing between-subject variability and increasing power. **Wilcoxon signed-rank tests** were conducted in parallel as non-parametric alternatives. All three hypotheses predict a specific direction and were therefore evaluated with **one-tailed tests**. Effect sizes were quantified using **Cohen’s d** for paired samples ($d = 0.2$: small, 0.5 : medium, 0.8 : large). The specific directional comparisons were:

- **Blind Spot:** $\text{REC}_{\text{post}} < \text{REC}_{\text{mid}}$
- **Snapshot:** $\text{LDI}_{\text{pre}} > \text{LDI}_{\text{mid}}$
- **Speed Bump:** $\text{RT}_{\text{post}} > \text{RT}_{\text{mid}}$

Multiple comparison correction. Running multiple tests inflates the probability of Type I error. Two correction methods were applied: **Bonferroni** (multiplies each p -value by the number of tests k ; controls the family-wise error rate conservatively) and **Benjamini–Hochberg** (controls the false discovery rate; less conservative, better suited for exploratory analyses). Both corrected p -values are reported alongside raw values throughout the Results.

3 Results

3.1 Overall Memory Performance

Table 1: Overall REC and LDI by experimental group.

Group	n	REC Mean	REC SD	LDI Mean	LDI SD
Item Only	56	0.685	0.187	0.374	0.236
Task Only	53	0.596	0.195	0.381	0.204
Both	51	0.641	0.190	0.307	0.251

All three groups exhibited positive REC and LDI (Table 1), confirming that participants could recognise studied items and discriminate lures. The Item Only group demonstrated the highest mean REC, while the Task Only group exhibited the lowest, suggesting encoding task type affects recognition quality. LDI values were comparable across groups.

3.2 Visualisations

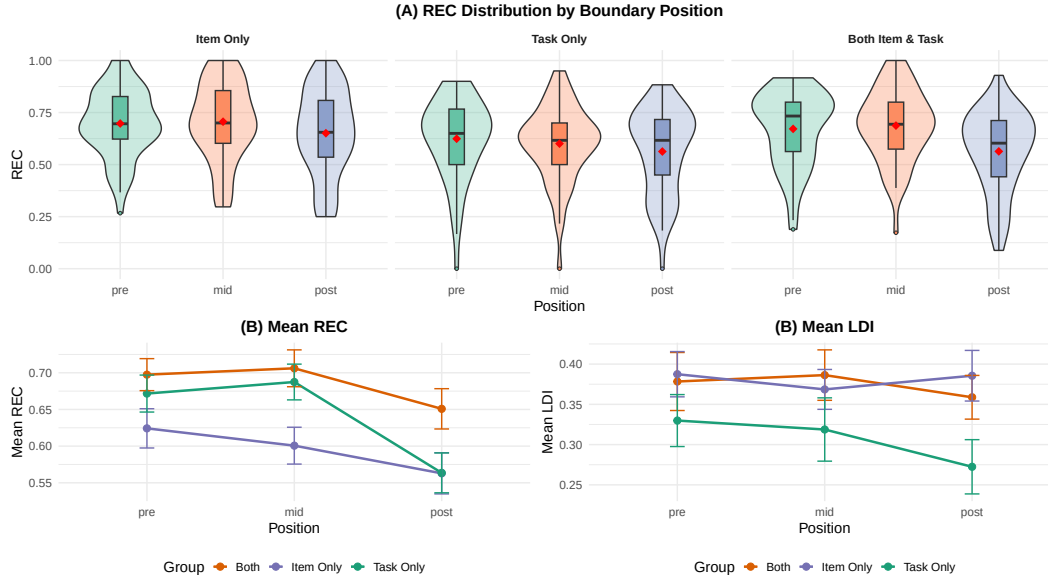


Figure 1: (A) REC by boundary position — violin plots with box plots and group means (red diamonds). A clear post-boundary drop is visible in the Both group. (B) Mean REC and LDI by position and group with ± 1 SE error bars. The Both group shows a clear mid-to-post REC decline; LDI is flat across positions.

Figure 1A displays the full REC distributions. In the Both Item & Task panel, the post-boundary violin and box are shifted downward relative to mid-event, with the mean (red diamond) sitting noticeably lower. The Item Only and Task Only groups show a subtler downward trend. Figure 1B makes the group trends easier to compare: the Both group shows a clear decline from mid-event to post-boundary for REC, while all three LDI lines are flat across positions, indicating no Snapshot effect at the descriptive level.

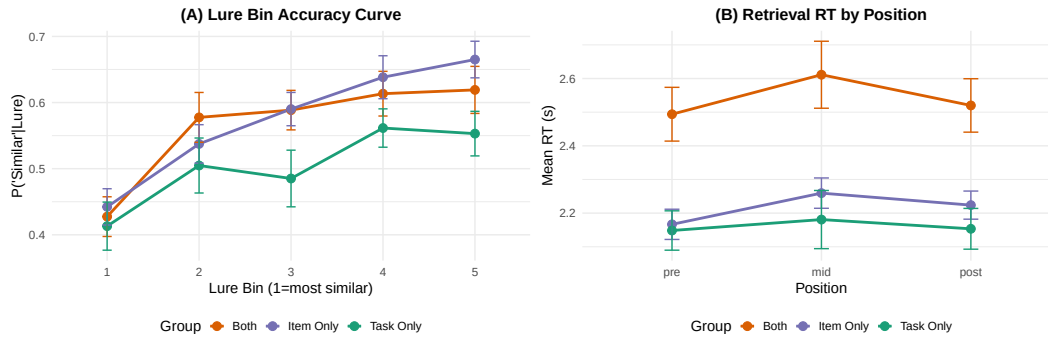


Figure 2: (A) Lure bin accuracy curve — all groups show the expected monotonic increase from Bin 1 (most similar, hardest) to Bin 5 (least similar, easiest), validating the MST. (B) Retrieval RT by boundary position — no systematic post-boundary slowing is observed.

Figure 2A presents the lure bin accuracy curve — a critical MST validation. All three groups

exhibit the expected monotonically increasing pattern, confirming participants engaged genuinely with the task and that the MST measured pattern separation. The overlapping curves show that encoding task type does not alter overall discrimination ability. Figure 2B shows mean retrieval RT by boundary position; no group exhibits systematic post-boundary slowing, previewing the null result for H3.

3.3 Normality Assessment

Table 2: Shapiro–Wilk normality tests by group and position.

Group	Position	REC W	REC p	LDI W	LDI p
Item Only	pre	0.977	0.367	0.975	0.295
Item Only	mid	0.961	0.064	0.983	0.633
Item Only	post	0.958	0.047	0.991	0.942
Task Only	pre	0.936	0.007	0.970	0.209
Task Only	mid	0.962	0.090	0.982	0.611
Task Only	post	0.939	0.009	0.973	0.271
Both	pre	0.924	0.003	0.968	0.179
Both	mid	0.978	0.461	0.968	0.178
Both	post	0.948	0.025	0.971	0.250

LDI did not depart from normality in any cell (all $p > 0.05$). REC showed significant departures in 5 of 9 cells, driven by extreme tail scores. However, parametric and non-parametric tests yielded identical conclusions throughout, confirming robustness to this violation.

3.4 Inferential Statistics

Table 3: Combined inferential results for all three hypotheses with multiple comparison corrections.

Hypothesis	Group	Mean						
		Diff	t	df	p (raw)	Cohen d	p (Bonf.)	p (BH)
H1: Blind Spot	Item Only	-0.055	-1.859	55	0.03419	-0.248	0.30772	0.11618
H1: Blind Spot	Task Only	-0.038	-1.801	52	0.03873	-0.247	0.34855	0.11618
H1: Blind Spot	Both	-0.124	-4.121	50	0.00007	-0.577	0.00064	0.00064
H2: Snapshot	Item Only	-0.008	-0.255	55	0.60000	-0.034	1.00000	0.77408
H2: Snapshot	Task Only	0.019	0.992	52	0.16279	0.136	1.00000	0.36627
H2: Snapshot	Both	0.011	0.379	50	0.35313	0.053	1.00000	0.63563

Hypothesis	Group	Mean Diff	t	df	p (raw)	Cohen d	p (Bonf.)	p (BH)
H3: Speed Bump	Item Only	-0.091	-0.758	55	0.77408	-0.101	1.00000	0.77408
H3: Speed Bump	Task Only	-0.036	-0.613	52	0.72880	-0.084	1.00000	0.77408
H3: Speed Bump	Both	-0.026	-0.265	50	0.60409	-0.037	1.00000	0.77408

Hypothesis 1 — Blind Spot (supported in the Both group). Recognition memory was significantly lower for post-boundary items than mid-event items in the Both Item & Task group ($t(50) = -4.12$, raw $p < .001$, $d = -0.58$), a medium-sized 12-percentage-point impairment that **survived both Bonferroni and BH corrections**. The Item Only and Task Only groups showed nominally significant trends in the same direction (raw $p = .034$ and $.039$), but these did not survive correction. Wilcoxon tests corroborated all parametric results.

Hypothesis 2 — Snapshot (not supported). No group showed a significant difference in LDI between pre-boundary and mid-event positions (all $p > .30$, all $|d| < 0.15$). Pre-boundary items did not exhibit enhanced lure discrimination, likely because boundaries were unpredictable and LDI estimates relied on relatively few lure trials per position per participant (9–11 trials).

Hypothesis 3 — Speed Bump (not supported). No group showed significantly slower retrieval RT for post-boundary items (all $p > .45$, all $|d| < 0.13$). This null result may reflect RT noise, cancellation of competing response tendencies, or the possibility that the speed bump operates during encoding rather than retrieval (Morse et al., 2023).

Additional finding — recognition gradient. Direct pre- vs. post-boundary comparisons revealed that pre-boundary items are recognised significantly better than post-boundary items in the Both group ($p = .0003$, $d = 0.55$) and the Task Only group ($p = .009$, $d = 0.37$), establishing a clear ordering: **pre** > **mid** > **post**. This gradient becomes more pronounced with increasing task complexity, supporting the interpretation that cognitively demanding encoding amplifies boundary effects.

4 Conclusion and Future Directions

This report examined event boundary effects on memory across three MST groups differing in encoding complexity. The central finding is strong support for the **Blind Spot hypothesis** in the Both Item & Task group: post-boundary items were recognised ~12 percentage points less accurately than mid-event items ($d = -0.58$, surviving stringent correction). The single-task groups exhibited the same trend but lacked power for corrected significance, suggesting the Blind Spot is a genuine phenomenon amplified under cognitive load. Neither the **Snapshot** nor the **Speed Bump** hypothesis was supported. The absence of a Snapshot

effect likely reflects the unpredictability of boundaries — without anticipation of the transition, the brain may not engage proactive stabilisation. The null Speed Bump may reflect RT noise or the possibility that slowing occurs at encoding rather than retrieval (Morse et al., 2023). These findings align with Event Segmentation Theory (Zacks et al., 2007): boundary-triggered context updating temporarily diverts encoding resources, and this disruption is magnified under cognitive load.

Report 2: Future Directions and Advanced Analysis. Report 2 will be an update of this initial report, moving from simple paired comparisons to advanced modeling that examines how different factors work together to influence memory. This next stage will include:

1. **Interaction Analysis (Two-way Repeated-Measures ANOVA):** In Report 1, we looked at groups separately. In Report 2, we will use ANOVA to formally test if the “memory drop” at boundaries is significantly more severe when encoding tasks are more complex. This helps us understand if a “busy brain” is more vulnerable to losing information during scene changes.
2. **Trial-Level Modeling (Mixed-Effects Logistic Regression):** Instead of using participant averages, we will analyze individual trials (every single image seen). This is a more powerful method that accounts for both individual differences in participants and the relative difficulty of specific images.
3. **Predictive Analysis (Generalized Linear Models - GLM):** We will investigate how factors like retrieval speed (Reaction Time) and how similar a fake image was to the real one (Lure Similarity) directly predict whether a participant remembers correctly.

Hypotheses for Report 2: *H4 (Task Interaction):* We expect the group doing the most complex task (Both Item & Task) to show a significantly steeper decline in memory at event boundaries compared to the simpler groups. *H5 (Similarity Gradient):* We predict that as the “Lure Bin” becomes more similar to the original image, participants will be significantly more likely to make errors (as predicted by our regression models). *H6 (Boundary Influence):* We expect our trial-level models to show that being at a “post-boundary” position is a significant predictor of memory error, even when controlling for how hard the specific image was.

5 Code Repository

The complete analysis code and reproducible pipeline used to generate the results and figures in this report is available at: https://github.com/Anurag-1205/MST_Simpletons

6 Author Contributions

The behavioral experiment itself was conducted as part of the course experiment pool. Our group participated in the experiment and performed statistical analysis on the dataset provided by the course instructors.

Work was divided among the team members based on the three hypotheses tested in the study.

Srushti Pekamwar — Hypothesis H1 (Blind Spot Effect). Analysis of recognition memory differences between post-boundary and mid-event items. Implementation of statistical tests and visualizations related to REC. Interpretation and documentation of the Blind Spot effect results.

Anurag Kaushal — Hypothesis H2 (Snapshot Effect). Analysis of lure discrimination differences between pre-boundary and mid-event items. Implementation of statistical tests and visualizations related to LDI. Interpretation and documentation of the Snapshot effect results.

Param Modi — Hypothesis H3 (Speed Bump Effect). Analysis of retrieval reaction time differences between post-boundary and mid-event items. Implementation of statistical tests and visualizations related to reaction time. Interpretation and documentation of the Speed Bump effect results.

All members collaboratively reviewed the statistical results and finalized the report.

7 References

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